Chamber Geometry Design for Small Bipropellant Liquid Rocket Engine

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1 Design

Fuel: ethanol.
Oxidizer: nitrous oxide.
Fuel to oxidizer mass ratio: 3 to 1 (Oxidizer to Fuel). This is fuel rich to reduce peak combustion temperatures.
Target max thrust: A) 30 N (6.74 lb). Estes D12-7 has max thrust of 29.7 N. B) 50 N (11.24 lb). This increases the nozzle throat diameter, lowering the contraction ratio. This thrust target is used for design candidate 6.
Chamber material: mild steel. This is cheaper than stainless steel.

The model is constructed using the following governing equations. Each equation is numbered and accompanied by a brief explanation that defines each variable and its corresponding MATLAB variable.

1. **Throat Diameter**

   Variable definition: D_t is the diameter at the nozzle throat, where the flow reaches Mach 1.

   $$D_t = \sqrt{\frac{4F}{\pi P_c C_F}} \tag{1}$$

   MATLAB: F = `F` (target thrust, N), P_c = `P_c` (chamber pressure, Pa), C_F = `C_F` (thrust coefficient, dimensionless), D_t = `D_t` (throat diameter, m).

2. **Throat Area**

   Variable definition: A_t is the cross-sectional area at the throat.

   $$A_t = \frac{\pi D_t^2}{4} \tag{2}$$

   MATLAB: D_t = `D_t` (throat diameter, m), A_t = `A_t` (throat area, m^2).

3. **Contraction Ratio**

   Variable definition: ϵ_c is the ratio of chamber area to throat area, representing how much the flow is accelerated into the throat.

$$\epsilon_c = \frac{\pi D_c^2}{4A_t} = \left(\frac{D_c}{D_t}\right)^2 \tag{3}$$

   MATLAB: D_c = `D_c` (chamber diameter, m), D_t = `D_t` (throat diameter, m), ϵ_c = `eps_c` (contraction ratio, dimensionless).

4. **Nozzle Length**

   Variable definition: L_n is the total length of the conical nozzle, including both the convergent and divergent sections, for a given half-angle θ. The total length is the sum of the convergent length (from chamber to throat) and the divergent length (from throat to exit).

$$L_n = \frac{D_c - D_t}{2\tan(\theta)} + \frac{D_e - D_t}{2\tan(\theta)} = \frac{D_c + D_e - 2D_t}{2\tan(\theta)} \tag{4}$$

   MATLAB: D_c = `D_c` (chamber diameter, m), D_t = `D_t` (throat diameter, m), D_e = `D_e` (exit diameter, m), θ = `Th` (nozzle half-angle, deg), L_n = `L_n` (nozzle length, m).

   Note: For the special case where the exit diameter equals the chamber diameter ($D_e = D_c$), this simplifies to:

$$L_n = \frac{D_c - D_t}{\tan(\theta)} \tag{4a}$$

5. **Characteristic Length**

   Variable definition: L_{ch} is the characteristic length, a measure of combustion residence time defined as chamber volume divided by throat area.

$$L_{ch} = L_c \epsilon_c \tag{5}$$

   MATLAB: L_c = `L_c` (chamber length, m), ϵ_c = `eps_c` (contraction ratio, dimensionless), L_{ch} = `L_ch` (characteristic length, m).

6. **Chamber L/D Ratio**

   Variable definition: $\mathrm{LD}_{\mathrm{rat}}$ is the length-to-diameter ratio of the cylindrical chamber, influencing combustion stability and residence time.

$$\mathrm{LD}_{\mathrm{rat}} = \frac{L_c}{D_c} \tag{6}$$

   MATLAB: L_c = `L_c` (chamber length, m), D_c = `D_c` (chamber diameter, m), $\mathrm{LD}_{\mathrm{rat}}$ = `LDrat` (L/D ratio, dimensionless).

7. **Expansion Ratio**

   Variable definition: ϵ is the ratio of nozzle exit area to throat area, determining how much the flow expands.

   $$\epsilon = \left(\frac{D_e}{D_t}\right)^2 \tag{7}$$

   MATLAB: D_e = `D_e` (exit diameter, m), D_t = `D_t` (throat diameter, m), ϵ = `eps` (expansion ratio, dimensionless).

8. **Exit Mach Number**

   Variable definition: M_e is the Mach number at the nozzle exit, solved iteratively from the isentropic area-Mach relation.

   $$\frac{A_e}{A_t} = \frac{1}{M_e}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M_e^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} \tag{8}$$

   MATLAB: A_e/A_t = `eps` (expansion ratio), γ = `ga` (specific heat ratio, dimensionless), M_e = `M_e` (exit Mach number, dimensionless).

9. **Exit Pressure**

   Variable definition: P_e is the pressure at the nozzle exit, calculated from the isentropic pressure relation.

   $$P_e = P_c\left(1+\frac{\gamma-1}{2}M_e^2\right)^{-\frac{\gamma}{\gamma-1}} \tag{9}$$

   MATLAB: P_c = `P_c` (chamber pressure, Pa), γ = `ga` (specific heat ratio), M_e = `M_e` (exit Mach number), P_e = `P_e` (exit pressure, Pa).

10. **Exit-to-Ambient Pressure Ratio**

    Variable definition: ϵ_{ep} is the ratio of exit pressure to ambient pressure. A value > 1 indicates under-expanded flow, which prevents flow separation and ensures stable operation.

    $$\epsilon_{ep} = \frac{P_e}{P_{amb}} \tag{10}$$

    MATLAB: P_e = `P_e` (exit pressure, Pa), P_{amb} = `P_amb` (ambient pressure, Pa), ϵ_{ep} = `eps_ep` (exit-to-ambient pressure ratio, dimensionless).

The following configuration inputs are:

$$
\begin{aligned}
\text{Max thrust } (F)&: 50 \text{ N} \\
\text{Chamber pressure } (P_c)&: 1,000 \text{ kPa} \\
\text{Thrust coefficient } (C_F)&: 1.3 \\
\text{Chamber diameter } &? \\
\text{Chamber length } &? \\
\text{Exit diameter } &? \\
\text{Nozzle angle } (\theta)&: 15^\circ \\
\text{Ambient pressure } (P_{amb})&: 85 \text{ kPa} \\
\text{Specific heat ratio } (\gamma)&: 1.27
\end{aligned}
$$

A model will be constructed:

$$
\begin{aligned}
f(F, P_c, C_F, D_c, L_c, \theta, D_e, P_{amb}, \gamma) &= \mathbf{H}, \\
\mathbf{H} &= [D_t, \epsilon_c, L_n, L^*, L/D, \epsilon, M_e, P_e, \epsilon_{ep}] \, .
\end{aligned}
$$

The model is constructed as follows:

$$D_t = \sqrt{\frac{4F}{\pi P_c C_F}}.$$

$$\epsilon_c = \frac{D_c^2 \pi}{4A_t},$$

where

$$A_t = \frac{\pi D_t^2}{4}.$$

And,

$$L_n = \frac{D_c - D_t}{\tan(\theta)}.$$

Also, the characteristic length and length-to-diameter ratio are:

$$L^* = L_c \epsilon_c, \quad L/D = \frac{L_c}{D_c}.$$

The expansion ratio is defined as:

$$\epsilon = \left(\frac{D_e}{D_t}\right)^2.$$

The exit Mach number M_e is found numerically by solving the area-Mach relation:

$$\epsilon = \frac{1}{M_e}\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2} M_e^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}.$$

The exit pressure and the exit-to-ambient pressure ratio are evaluated using isentropic flow relations:

$$P_e = P_c \left(1 + \frac{\gamma - 1}{2} M_e^2\right)^{-\frac{\gamma}{\gamma-1}},$$

$$\epsilon_{ep} = \frac{P_e}{P_{amb}}.$$

The output is $\mathbf{H} = [D_t, \epsilon_c, L_n, L^*, L/D, \epsilon, M_e, P_e, \epsilon_{ep}]$.

Recommended Candidate Geometries

All candidates use the fixed inputs:

$$F = 30 \text{ N}, \quad P_c = 1000 \text{ kPa}, \quad C_F = 1.3, \quad \theta = 15^\circ$$

which yield $D_t = 5.42$ mm (drill 5.5 mm) for every case.

Candidate 1 (20mm Diameter)

Inputs	**Values**
Chamber diameter D_c	20.0 mm (0.020 m)
Chamber length L_c	80.0 mm (0.080 m)

Outputs	**Values**
Throat diameter D_t	5.42 mm
Contraction ratio ϵ_c	13.6
Nozzle length L_n	54.4 mm
Characteristic length L^*	1.09 m
Exit diameter $D_e = D_c$	20.0 mm

Candidate 2 (18mm Diameter)

Inputs	**Values**
Chamber diameter D_c	18.0 mm (0.018 m)
Chamber length L_c	90.0 mm (0.090 m)

Outputs	**Values**
Throat diameter D_t	5.42 mm
Contraction ratio ϵ_c	11.0
Nozzle length L_n	46.9 mm
Characteristic length L^*	0.99 m
Exit diameter $D_e = D_c$	18.0 mm

Candidate 3 (22mm Diameter)

Inputs	Values
Chamber diameter D_c	22.0 mm (0.022 m)
Chamber length L_c	60.0 mm (0.060 m)

Outputs	Values
Throat diameter D_t	5.42 mm
Contraction ratio ϵ_c	16.5
Nozzle length L_n	61.8 mm
Characteristic length L^*	0.99 m
Exit diameter $D_e = D_c$	22.0 mm

Candidate 4 (20mm Diameter, 100mm Length)

Inputs	Values
Chamber diameter D_c	20.0 mm (0.020 m)
Chamber length L_c	100.0 mm (0.100 m)

Outputs	Values
Throat diameter D_t	5.42 mm
Contraction ratio ϵ_c	13.6
Nozzle length L_n	54.4 mm
Characteristic length L^*	1.36 m
Exit diameter $D_e = D_c$	20.0 mm
L/D ratio	5.0

Candidate 5 (20mm Diameter, 60mm Length, 50N Thrust)

Inputs	Values
Chamber diameter D_c	20.0 mm (0.020 m)
Chamber length L_c	60.0 mm (0.060 m)
Max thrust F	50.0 N
Chamber pressure P_c	1000.0 kPa
Thrust coefficient C_F	1.300
Nozzle angle θ	15.0°

Outputs	Values
Throat diameter D_t	6.998 mm
Contraction ratio ϵ_c	8.168
Nozzle length L_n	48.524 mm
Characteristic length L^*	0.490 m
Exit diameter $D_e = D_c$	20.0 mm
L/D ratio	3.000

Justification: The target thrust is increased from 30 N to 50 N, which increases the throat diameter from 5.42 mm to 6.998 mm and lowers the contraction ratio from 13.6 to 8.168. This falls within the recommended range (ϵ_c = 8–10) for small bipropellant engines operating at sea level. The chamber length is decreased from 100 mm to 60 mm, providing a L/D ratio of 3.0. The characteristic length is $L^* = 0.490$ m.

2 Candidate 6

Model Input Summary

Symbol	Physical Meaning	MATLAB Variable	Value
F	Target thrust	`F`	50 N
P_c	Chamber pressure	`P_c`	1,000 kPa
C_F	Thrust coefficient	`C_F`	1.3
D_c	Chamber diameter	`D_c`	20 mm
L_c	Chamber length	`L_c`	100 mm
θ	Nozzle half-angle	`Th`	15 deg
D_e	Exit diameter	`D_e`	10 mm
P_{amb}	Ambient pressure	`P_amb`	85 kPa
γ	Specific heat ratio	`ga`	1.27

Model Output Summary

Symbol	Physical Meaning	MATLAB Variable	Value
D_t	Throat diameter	`D_t`	6.998 mm
ϵ_c	Contraction ratio	`eps_c`	8.168
L_n	Nozzle length	`L_n`	29.86 mm
L_{ch}	Characteristic length	`L_ch`	0.817 m
$\mathrm{LD}_{\mathrm{rat}}$	L/D ratio	`LDrat`	5.0
ϵ	Expansion ratio	`eps`	2.042
M_e	Exit Mach number	`M_e`	2.124
P_e	Exit pressure	`P_e`	106.727 kPa
ϵ_{ep}	Exit-to-ambient pressure ratio	`eps_ep`	1.256

Design Criteria Check

Characteristic length: $L_{ch} = 0.817$ m (recommended range: 0.75–1.25 m).
L/D ratio: $\mathrm{LD}_{\mathrm{rat}} = 5.0$ (recommended range: 4–6).
Contraction ratio: $\epsilon_c = 8.168$ (recommended range: 2.5–10).
Exit pressure ratio: $\epsilon_{ep} = 1.256$ (recommended range: 1.0–1.5).